

Notes: Pool, Stoffman & Yonker (JF, 2015)
The People in Your Neighborhood: Social Interactions and Mutual Fund
Portfolios

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Mutual fund holdings and trades	4
3.2	Manager residential addresses	4
3.3	Normalized driving distance and neighbor definition	4
3.4	Overlap measures	4
3.5	Control variables	5
4	Models and empirical specifications	5
4.1	Baseline holdings-overlap regression	5
4.2	Trade-overlap regressions	5
4.3	Extensive and intensive margins	5
4.4	Mechanism regressions	6
4.5	Performance tests	6
5	Identification and empirical design	6
5.1	Core identification challenge	6
5.2	Why residential proximity matters	6
5.3	Disentangling social interaction from community effects	6
5.4	Falsification and mechanism checks	7
5.5	Performance as a mechanism test	7
6	Tables and figures	7
6.1	Figure 1: Cumulative distributions of neighbor proximity; Figure 2: City clusters and neighbor pairs	7
6.2	Table I: Summary Statistics	8
6.3	Table II: Social Interactions and Mutual Fund Holdings	8
6.4	Table III: Social Interactions and Mutual Fund Trades; Table IV: Extensive and Intensive Margin Overlap	10
6.5	Table V: Social Interactions and Manager Characteristics; Table VI: Social Interactions versus Preferences: Neighbor Relocations	10
6.6	Figure 3: Portfolio overlap and manager distance; Figure 4: Portfolio overlap and neighbor tenure	11
6.7	Table VII: Social Interactions versus Preferences: Religion and Political Variables; Table VIII: Social Interactions versus Preferences: Indirect Neighbors	12

6.8	Table IX: The Performance Effect of Social Interactions: Holdings; Table X: The Performance Effect of Social Interactions: Trades	14
6.9	Table XI: Social Interactions and Stock Characteristics; Table XII: Social Interactions and Hard-to-Research Stocks	14
6.10	Figure 5: Bootstrap Analysis	15
7	Short summary	16
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic mechanism	17
8.3	Why the identification is persuasive	17
8.4	Limitations	17
8.5	Takeaway	17

1 Big picture

- **Question.** Do professional mutual fund managers exchange value-relevant investment ideas through local person-to-person social interactions?
- **Core idea.** Managers who live near one another are more likely to meet, become acquainted, and communicate. If social transmission matters, funds run by residential neighbors should hold and trade more similar stocks than other funds in the same city or media market.
- **Identification wedge.** The paper separates *social interactions* from broad *community effects*. Same city or same media market captures shared local media, local roadshows, and local job-market effects; residential proximity within that city captures a much narrower channel of likely person-to-person interaction.
- **Main finding.** Neighboring managers have significantly higher overlap in holdings and trades than non-neighboring managers, even with controls for city, media market, same fund family, fund style, size, and family relationships.
- **Mechanism evidence.** The effect is stronger when managers share ethnicity, grows with neighbor tenure, appears among indirect neighbors, and is not explained by future-neighbor placebo effects or preference proxies.
- **Value evidence.** Stocks bought by neighboring managers and stocks sold by neighboring managers generate return patterns consistent with valuable information transmission, especially for hard-to-research stocks.
- **Takeaway.** Social networks do not only shape retail investor behavior. They also affect the investment decisions and information production of professional money managers.

2 Incremental contribution of the paper

This paper contributes to the literature on social interactions, delegated asset management, and local information in four main ways.

1. **It identifies social interactions inside a professional investor population.** Prior work shows that investors in the same city or school network behave similarly. This paper goes further by using fund managers' residential address histories to isolate likely social contact among nearby managers.
2. **It separates social effects from community effects.** The key comparison is not simply local versus nonlocal. It is neighbor managers versus same-city or same-media-market managers who are not neighbors. This distinction helps rule out shared media markets, local corporate roadshows, and local labor-market segmentation.
3. **It links social interaction to both portfolio choices and information value.** The paper does not stop at documenting correlated holdings. It studies trades, extensive and intensive margins, performance of neighbor buy/sell portfolios, and the concentration of effects in hard-to-research stocks.
4. **It develops a set of falsification and mechanism tests.** Future-neighbor, past-neighbor, tenure, ethnic similarity, religious-area, indirect-neighbor, and bootstrap tests collectively build the case that the relation reflects information transmission rather than stable tastes or mechanical sorting.

Interpretation: The incremental contribution is the measurement of an informal peer network among professionals and the demonstration that this network transmits economically valuable stock ideas.

3 Data and measurement

3.1 Mutual fund holdings and trades

The paper begins with actively managed U.S. mutual funds from 1996Q1 through 2010Q4. Holdings are observed quarterly, and manager identities are linked to funds. The final analysis uses quarterly fund-pair observations, creating a large panel of potential social connections among managers.

- The sample averages about 412 funds per quarter.
- It covers more than 4 million quarterly fund-pair observations in the baseline summary statistics.
- Trades are inferred from quarterly changes in holdings: purchases are increases or new positions; sales are decreases or liquidations.

Interpretation: The unit of observation is usually a fund pair in a quarter. This design asks whether two funds managed by socially connected managers make more similar portfolio decisions in the same quarter.

3.2 Manager residential addresses

The paper collects residential address histories for mutual fund managers from public records. These records include owned and rented properties and allow the authors to locate managers' residences at the beginning of each quarter.

- Managers are linked to address histories through name, age, and location searches.
- Distances are calculated from the latitude and longitude of managers' residential locations.
- The authors identify 2,042 managers in the address data out of 4,622 managers in the initial sample.

Interpretation: The novelty of the data is that it moves the network measure from institutional affiliation to residential proximity, which is a more direct proxy for possible social contact.

3.3 Normalized driving distance and neighbor definition

Residential proximity is adjusted for population density. A mile in Manhattan is socially different from a mile in a suburb, so the paper normalizes driving distance by local household density:

$$NDD_{ij} = DrivingDistance_{ij} \times \frac{\max(hhdens_i, hhdens_j)}{\text{median}(hhdens)}.$$

Managers are classified as neighbors if their normalized driving distance is no more than 2.6 normalized miles.

Interpretation: This density adjustment is important. It makes the neighbor definition comparable across dense cities and low-density suburbs. A very short physical distance in Manhattan can be socially similar to several miles in a suburb.

3.4 Overlap measures

For holdings, the overlap between funds i and j in quarter t is:

$$PortOverlap_{ijt} = \sum_{k \in H_t} \min(w_{ikt}, w_{jkt}),$$

where w_{ikt} is fund i 's portfolio weight in stock k .

For trades, the paper constructs purchase and sale overlap. These measures ask whether two funds buy or sell the same stocks in the same quarter, with separate measures for extensive overlap and intensive overlap.

Interpretation: Holdings overlap is a stock of common portfolio positions. Trade overlap is closer to the information-flow mechanism because it captures synchronized active decisions.

3.5 Control variables

The paper controls for many reasons why two funds might look similar without social interaction:

- same fund family;
- same management city and same media market;
- fund style categories: growth, blend, value, small-cap, mid-cap, large-cap;
- fund size differences and average fund size;
- whether at least one manager manages both funds;
- whether the fund pair has common or related manager structures.

Interpretation: These controls are meant to prevent the neighbor coefficient from picking up fund strategy, institutional family, or broad local information channels.

4 Models and empirical specifications

4.1 Baseline holdings-overlap regression

The baseline regression is:

$$Overlap_{ijt} = \alpha + \beta Neighbor_{ijt} + \delta SameMFCity_{ijt} + \gamma SameMediaMkt_{ijt} + \Gamma' Controls_{ijt} + \varepsilon_{ijt}.$$

Interpretation: The coefficient β measures the extra overlap of neighbor-managed funds relative to otherwise similar fund pairs, including pairs in the same city or media market but not in the same neighborhood.

4.2 Trade-overlap regressions

The trade tests replace holdings overlap with purchase and sale overlap:

$$BuyOverlap_{ijt} = \alpha + \beta Neighbor_{ijt} + \delta SameMFCity_{ijt} + \gamma SameMediaMkt_{ijt} + \Gamma' Controls_{ijt} + \varepsilon_{ijt},$$

$$SaleOverlap_{ijt} = \alpha + \beta Neighbor_{ijt} + \delta SameMFCity_{ijt} + \gamma SameMediaMkt_{ijt} + \Gamma' Controls_{ijt} + \varepsilon_{ijt}.$$

Interpretation: A positive neighbor coefficient for purchases is especially consistent with managers sharing new stock ideas. A positive coefficient for sales is consistent with sharing negative information or common exit decisions.

4.3 Extensive and intensive margins

The paper decomposes trade overlap into extensive and intensive margins:

- **Extensive overlap:** whether both funds trade the same stock.
- **Intensive overlap:** whether both funds change an existing common position.

Interpretation: Extensive overlap is closer to idea generation. Intensive overlap can also reflect portfolio rebalancing among stocks already held by both funds.

4.4 Mechanism regressions

The paper estimates variants that interact *Neighbor* with manager characteristics:

$$Overlap_{ijt} = \alpha + \beta Neighbor_{ijt} + \theta Neighbor_{ijt} \times Z_{ij} + \Gamma' Controls_{ijt} + \varepsilon_{ijt},$$

where Z_{ij} includes same ethnicity, similar age, same college, experience, or fund-family links.

Interpretation: If social transmission is easier among managers with similar backgrounds, θ should be positive for characteristics that increase trust or communication intensity.

4.5 Performance tests

The paper forms portfolios of stocks that funds buy or sell with neighboring managers and compares their risk-adjusted returns to non-neighbor trades and benchmark returns. The key long-short idea is:

$$Return^{LS} = Return(NeighborBuys) - Return(NeighborSells).$$

Interpretation: If neighbor trades transmit valuable information, stocks bought by neighbors should outperform stocks sold by neighbors, especially when the stocks are hard to research.

5 Identification and empirical design

5.1 Core identification challenge

The empirical challenge is to determine whether similar fund choices reflect direct social interactions or broader community effects. Managers in the same city share many features: local media, local corporate visits, same regional job market, and local tastes. The paper’s solution is to compare truly nearby residential neighbors to managers in the same city or media market who are not neighbors.

Interpretation: The identifying variation is within-city residential proximity. The paper asks whether moving from same local environment to same neighborhood adds explanatory power.

5.2 Why residential proximity matters

The paper argues that managers who live near one another have more opportunities for contact: schools, commutes, places of worship, neighborhood events, local clubs, and repeated casual encounters. These encounters may not be the channel through which ideas are exchanged, but they increase the likelihood of forming relationships that later support deliberate communication.

Interpretation: The neighbor dummy is a proxy for relationship formation, not a claim that investment tips are exchanged during random street encounters.

5.3 Disentangling social interaction from community effects

The baseline specification includes both *SameMFCity* and *SameMediaMkt*. This means *Neighbor* is interpreted relative to fund pairs already allowed to share a city or media market.

Neighbor effect = extra overlap within the neighborhood beyond same-city and same-media-market effects.

Interpretation: This is the central design strength. If only broad city-level information explained the results, same-city and same-media-market controls would absorb the effect.

5.4 Falsification and mechanism checks

The paper adds several tests that strengthen the social interpretation:

- **Future neighbors.** Managers who will become neighbors later should not overlap before becoming neighbors if residential proximity is not yet creating social interaction.
- **Past neighbors.** Former neighbors may continue to share information if relationships persist after relocation.
- **Neighbor tenure.** Effects should grow with years spent as neighbors if relationships take time to form.
- **Ethnicity and religion-area tests.** Shared background should strengthen communication if trust and social affinity matter.
- **Indirect neighbors.** Information can travel through a chain of social connections.
- **Bootstrap.** Randomly assigning neighbors inside media markets should not reproduce the observed coefficient.

Interpretation: These tests target different alternative explanations. Future-neighbor tests address stable preferences, tenure tests address timing, indirect-neighbor tests address network diffusion, and bootstrap tests address random spatial correlation.

5.5 Performance as a mechanism test

The performance tests are not merely return exercises. They ask whether overlapping neighbor trades contain information not already reflected in benchmark characteristics.

Interpretation: If neighbor overlap were only caused by style preferences or herding, there would be little reason for neighbor buy-sell portfolios to earn abnormal returns, especially in opaque stocks. Positive risk-adjusted returns support an information-transmission interpretation.

6 Tables and figures

6.1 Figure 1: Cumulative distributions of neighbor proximity; Figure 2: City clusters and neighbor pairs

Read: Figure 1 shows that neighbors are physically close even after density adjustment: the median driving distance between neighbor pairs is about two miles, and the median number of households within that radius is about 1,874. Figure 2 maps city clusters and neighbor pairs, showing that the network is geographically broad and not limited to one finance center.

Economic message: The figures validate the neighbor construction. The distance threshold is tight enough to proxy for real social proximity, while the map shows that the test is not simply a New York or Boston phenomenon.

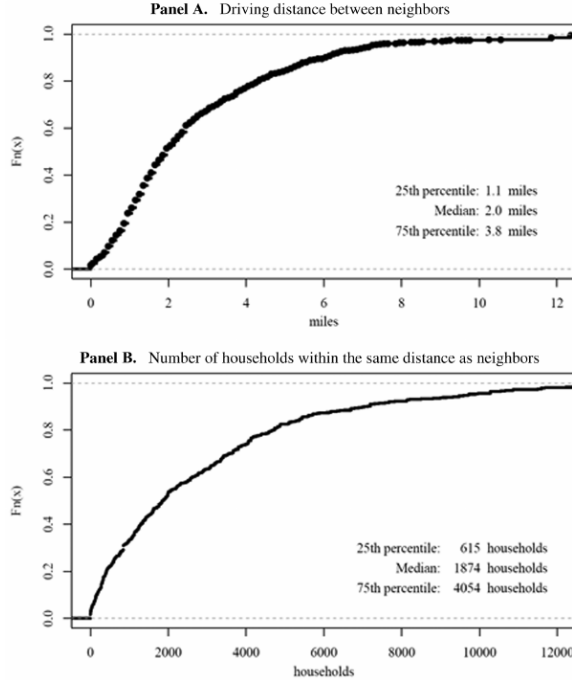


Figure 1. Cumulative distributions of neighbor proximity. The figure shows cumulative distributions of driving distances (Panel A) and number of households within the same linear distance (Panel B) for neighbor pairs. Managers in a pair are classified as neighbors if they live within 2.6 normalized miles of one another. The number of households within the same distance as manager pair (i, j) is calculated as $\pi \times \left(\frac{DD_{i,j}}{1.96}\right)^2 \times \max(HHD_i, HHD_j)$, where $DD_{i,j}$ is the driving distance between the residences of the managers. HHD_i is the household density in manager i 's

(a) Figure 1: Cumulative distributions of neighbor proximity. The two panels show physical closeness and the number of households inside the comparable radius.

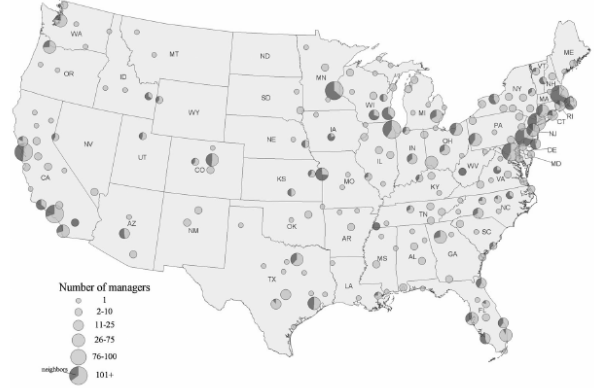


Figure 2. City clusters and neighbor pairs. The figure shows the geographic dispersion of fund managers. We use a clustering algorithm to group nearby zip codes into cities, as explained in the text. The algorithm stops joining clusters that are more than 50 miles apart. The size of the plotted circle is determined by the number of unique managers who have a residence in the city at some point during our sample period, while the shaded pie slice indicates the proportion of managers who are classified as neighbors with at least one other manager at some point during

(b) Figure 2: City clusters and neighbor pairs. The map shows the geographic distribution of fund managers and neighbor classifications.

6.2 Table I: Summary Statistics

Read: Table I compares all fund pairs, same-media-market non-neighbors, and true neighbors. Neighbor pairs are rare, but they exhibit higher overlap in holdings and trades. The table also shows that neighbors are more likely to share fund-family or style characteristics, which motivates the rich controls in later regressions.

Interpretation: The table is important because raw neighbor pairs are not randomly assigned. They differ from other fund pairs on some observables. The later regressions therefore need to condition on same media market, same city, same family, fund styles, and size measures.

6.3 Table II: Social Interactions and Mutual Fund Holdings

Read: The coefficient on *Neighbors* is positive and significant across specifications. In the full specification, neighbor funds hold more similar portfolios than non-neighbor funds even after controlling for same city, same media market, same fund family, style, and size. The single-manager specification produces an especially large estimate, implying economically meaningful extra overlap.

Interpretation: Table II is the baseline evidence that social proximity matters for portfolio choices. The comparison with *SameMFCity* and *SameMediaMkt* shows that the effect is not merely a broad local information-market effect.

Table I
Summary Statistics

The table reports means of the overlap measures, measures of social interactions, and control variables for the full sample of observations, observations for which managers live within 2.6 normalized miles of one another (*Neighbors*), observations for which managers live within 50 miles of one another (*SameMediaMit*), and observations for which managers live more than 50 miles apart. Variable names appear in *italics*. The sample includes 4.1 million quarterly fund-pair observations from the first quarter of 1996 through the fourth quarter of 2010. Manager-specific control variables are constructed at the manager-pair level. The sample includes 8.8 million manager pairs from the first quarter of 1996 through the fourth quarter of 2010. Means in column (3) are compared to (2), and in column (4) are compared to column (3). Significance levels for tests of difference in these means are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively. Standard errors are two-way clustered by each fund in the pair for these tests. * denotes unconditional means.

	(1)	(2)	(3)	(4)	
	Mean when				
<i>SameMediaMit</i> =	*	0	1	1	
<i>Neighbors</i> =	*	0	0	1	<i>N</i> (000's)
Dependent variables					
% of overlapping holdings	8.02	7.91	8.86 ^a	11.06 ^a	4,090
% of overlapping purchases	8.48	8.34	9.52 ^a	11.68 ^a	3,370
% of overlapping sales	6.67	6.57	7.52 ^a	8.99 ^a	3,370
Independent variables of interest					
Driving dist. between homes ≤ 1 mile (<i>DistBetween1Mile</i>)	0.003				4,090
Norm. driving dist. between homes ≤ 2.6 miles (<i>Neighbors</i>)	0.003				4,090
Baseline control variables					
Zip code distance between homes ≤ 50 miles (<i>SameMediaMit</i>)	0.10				4,090
Funds are located in the same city (<i>SameMPCity</i>)	0.07	0.04	0.22 ^a	0.41 ^a	4,090
Funds are in the same family (<i>SameFundFam</i>)	0.01	0.01	0.05 ^a	0.11 ^a	4,090
Mgrs. mng. at least 1 other fund together (<i>MngOtherFundTogether</i>)	0.00	0.00	0.01 ^a	0.04 ^a	4,090
Funds have at least 1 mgr. in common (<i>CommonMgr</i>)	0.00	0.00	0.01 ^a	0.05 ^a	4,090
Funds are both small-cap (<i>BothSmallCap</i>)	0.07	0.06	0.07	0.06	4,090
Funds are both mid-cap (<i>BothMidCap</i>)	0.06	0.06	0.07	0.04	4,090
Funds are both large-cap (<i>BothLargeCap</i>)	0.41	0.41	0.43	0.47	4,090
Funds are both value (<i>BothValue</i>)	0.05	0.05	0.06 ^a	0.06	4,090

(Continued)

Table I: Summary statistics, part 1. The table reports means by neighbor and media-market status and motivates the regression controls.

Table I—Continued

	(1)	(2)	(3)	(4)	
	Mean when				
<i>SameMediaMit</i> =	*	0	1	1	
<i>Neighbors</i> =	*	0	0	1	<i>N</i> (000's)
Funds are both growth (<i>BothGrowth</i>)	0.23	0.24	0.21 ^a	0.20 ^a	4,090
Funds are both blend (<i>BothBlend</i>)	0.09	0.09	0.10	0.08 ^a	4,090
Difference in TNA-based quintiles between fund pairs (<i>TNAQuintDiff</i>)	1.59	1.59	1.64 ^b	1.53	4,090
Average TNA-based quintile of fund pair (<i>TNAQuintAvg</i>)	3.02	3.01	3.13 ^a	3.23 ^a	4,090
Manager-specific control variables					
Number of years managers have been neighbors (<i>NeighborTenure</i>)	6.84				15
Both mgrs. reside in religious areas (<i>BothReligiousAreas</i>)	0.27	0.23	0.09 ^a	0.72	8,846
Both mgrs. reside in Jewish areas (<i>BothJewishAreas</i>)	0.09	0.07	0.27 ^a	0.38 ^a	8,846
Both mgrs. reside in Catholic areas (<i>BothCatholicAreas</i>)	0.09	0.07	0.27 ^a	0.53 ^a	8,846
Both mgrs. reside in high home value areas (<i>BothHighHomeValueAreas</i>)	0.07	0.06	0.24 ^a	0.43 ^a	8,846
Both mgrs. reside in low home value areas (<i>BothLowHomeValueAreas</i>)	0.06	0.06	0.02 ^a	0.03 ^a	8,846
Ratio of the current value of mgrs' prim. residences (<i>HomePriceRatio</i>)	0.44	0.44	0.50 ^a	0.58 ^a	6,615
Ratio of the log size of mgrs' prim. residences (<i>LogSizeRatio</i>)	0.37	0.37	0.37	0.49 ^a	4,666
Ratio of the home age of mgrs' prim. residences (<i>HomeAgeRatio</i>)	0.37	0.37	0.39 ^a	0.39	5,127
Difference in mgrs' ages ≤ 10 years (<i>SimilarAge</i>)	0.52	0.51	0.53 ^a	0.58 ^a	8,846
Mgrs. of the same minority ethnicity (<i>SameEthnicity</i>)	0.01	0.01	0.01 ^a	0.02	8,846
Mgrs. graduated from the same college (<i>SameCollege</i>)	0.02	0.01	0.04 ^a	0.10 ^a	6,114
Both mgrs. are experienced (<i>BothExp</i>)	0.27	0.27	0.22 ^a	0.34 ^a	7,090
Both mgrs. are inexperienced (<i>BothInexp</i>)	0.25	0.25	0.29 ^a	0.23 ^a	7,090

Social Interactions and Mutual Fund Portfolios

2687

Table I: Summary statistics, part 2. The continuation reports style, family, and manager-pair characteristics.

Table II
Social Interactions and Mutual Fund Holdings

The table reports the coefficient estimates and standard errors from the OLS estimation of various forms of the regression equation

$$Overlap_{i,j} = \alpha + \beta Neighbors_{i,j} + \delta SameMPCity_{i,j} + \gamma SameMediaMit_{i,j} + \tau' Controls_{i,j} + \epsilon_{i,j},$$

where $Overlap_{i,j}$ is the overlap in holdings of fund i and fund j during quarter t and is defined as in equation (1). The variable of interest is $Neighbors_{i,j}$, a dummy variable that is one if at least one manager from fund i lives within 2.6 normalized miles of at least one manager from fund j during quarter t . $SameMPCity_{i,j}$ is a dummy variable that is one if funds i and j are headquartered in the same city. $SameMediaMit_{i,j}$ is a dummy variable that is one if at least one pair of managers for a given fund pair live within 50 miles of one another, and $Controls_{i,j}$ is a vector of relevant control variables. In column 2 the variable of interest is a dummy variable that is one if the raw driving distance between the residences of at least one pair of managers for a given fund pair is one mile or less (*DistBetween1Mile*). Controls include a dummy variable that is one if funds i and j are headquartered in the same city (*SameMPCity*), a dummy variable that is one if at least one pair of managers for a given fund pair live within 50 miles of one another (*SameMediaMit*), dummy variables that are one if funds i and j match on Morningstar size or value/growth categories (*BothSmallCap*, *BothMidCap*, *BothLargeCap*, *BothValue*, *BothGrowth*, *BothBlend*), an indicator variable that is one if funds i and j are from the same mutual fund family (*SameFundFam*), an indicator variable that is one if at least one pair of managers from funds i and j manage at least one other fund together (*MngOtherFundTogether*), an indicator variable that is one if funds i and j have at least one manager in common (*CommonMgr*), the absolute value of the difference between the total net asset (TNA)-based quintiles of the funds in the pair (*TNAQuintDiff*), and the average TNA quintiles of the funds in the pair (*TNAQuintAvg*), where TNA quintiles are computed quarterly. The sample includes 4.1 million quarterly fund-pair observations from the first quarter of 1996 to the fourth quarter of 2010. In column 5, the sample is limited to fund pairs with no managers in common during quarter t . Column 6 includes fund pairs within the same fund family. In column 7 the results are presented for the sample of funds where both funds in the pair are managed by a single manager. Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

Sample	Full (1)	Full (2)	Full (3)	Full (4)	No Mgrs. Common (5)	Different Families (6)	Single Mgr. Pairs (7)
<i>Neighbors</i>			1.29 ^a	0.99 ^a	0.99 ^a	0.88 ^a	2.24 ^a
<i>DistBetween1Mile</i>		0.84 ^b					
<i>SameMPCity</i>	0.36 ^a	0.35 ^a	0.34 ^b	0.20	0.19	0.20	0.11
<i>SameMediaMit</i>	0.16 ^a	0.16	0.15	0.15	0.15	0.15	0.34
<i>SameFundFam</i>	3.64 ^a	3.64 ^a	3.63 ^a	3.63 ^a	3.70 ^a	3.63 ^a	6.10 ^a
<i>MngOtherFundTogether</i>	0.38	0.38	0.37	0.37	0.38	0.38	0.82
	1.79 ^a	1.78 ^a	1.79 ^a	1.77 ^a	1.55 ^a	0.88	0.88
	(0.46)	(0.46)	(0.46)	(0.46)	(0.46)	(0.46)	(1.14)

(Continued)

Table II: Holdings overlap, part 1. The table estimates the baseline neighbor effect on portfolio overlap.

Table II—Continued

	Full (1)	Full (2)	Full (3)	Full (4)	No Mgrs. Common (5)	Different Families (6)	Single Mgr. Pairs (7)
<i>CommonMgr</i>	9.16 ^a	9.12 ^a	9.07 ^a	8.99 ^a			
<i>BothSmallCap</i>	0.79	0.79	1.21 ^a	1.21 ^a	1.19 ^a	1.18 ^a	1.19 ^a
<i>BothMidCap</i>	0.16	0.16	0.16	0.16	0.16	0.16	0.28
<i>BothLargeCap</i>	1.78 ^a	1.78 ^a	1.78 ^a	1.79 ^a	1.77 ^a	1.79 ^a	1.21 ^a
<i>BothValue</i>	0.20	0.20	0.20	0.20	0.20	0.20	0.30
<i>BothGrowth</i>	12.06 ^a	12.06 ^a	12.06 ^a	12.06 ^a	12.05 ^a	12.02 ^a	12.04 ^a
<i>BothBlend</i>	0.36	0.36	0.36	0.36	0.36	0.36	0.64
<i>BothExp</i>	2.84 ^a	2.84 ^a	2.84 ^a	2.84 ^a	2.84 ^a	2.84 ^a	2.17 ^a
<i>BothInexp</i>	0.42	0.42	0.42	0.42	0.42	0.42	0.88
<i>BothReligiousAreas</i>	3.44 ^a	3.44 ^a	3.44 ^a	3.44 ^a	3.43 ^a	3.43 ^a	3.31 ^a
<i>BothJewishAreas</i>	0.24	0.24	0.24	0.24	0.24	0.24	0.40
<i>BothCatholicAreas</i>	2.35 ^a	2.35 ^a	2.35 ^a	2.34 ^a	2.34 ^a	2.34 ^a	1.70 ^a
<i>BothHighHomeValueAreas</i>	0.33	0.33	0.33	0.33	0.33	0.33	0.40
<i>BothLowHomeValueAreas</i>	-0.18 ^a	-0.18 ^a	-0.18 ^a	-0.18 ^a	-0.18 ^a	-0.17 ^a	-0.22 ^a
<i>TNAQuintDiff</i>	0.03	0.03	0.03	0.03	0.03	0.03	0.06
<i>TNAQuintAvg</i>	0.66 ^a	0.66 ^a	0.66 ^a	0.66 ^a	0.66 ^a	0.66 ^a	0.73 ^a
<i>Constant</i>	-0.11	-0.11	-0.11	-0.12	-0.12	-0.08	-0.19
<i>Adj R²</i>	0.30	0.30	0.30	0.30	0.30	0.30	0.56
<i>N</i> (Observations)	4,090	4,090	4,090	4,090	4,084	4,081	472

Social Interactions and Mutual Fund Holdings

2686

Table II: Holdings overlap, part 2. The continuation reports additional specifications and controls.

6.4 Table III: Social Interactions and Mutual Fund Trades; Table IV: Extensive and Intensive Margin Overlap

Read: Table III shows that neighbors have greater overlap in buys and, to a lesser extent, sales. The purchase coefficients are economically larger and more robust. Table IV decomposes trade overlap and shows that the neighbor effect appears in both extensive and intensive margins, with especially strong evidence for common purchase decisions.

Interpretation: The trade results sharpen the mechanism. Holdings overlap could reflect slow-moving style or preference similarity; trade overlap is more naturally connected to timely information sharing.

Table III
Social Interactions and Mutual Fund Trades

The table reports the coefficient estimates and standard errors from OLS estimation of purchase and sale overlap on *Neighbors* and the control variables defined in Table II. The dependent variable for the regression results displayed in columns 1 through 3 (4 through 6) is the percentage of overlapping stock purchases (sales) between a given pair of funds during the quarter. These dependent variables are defined in equations (2) and (3), respectively. The sample includes 3.4 million quarterly fund-pair observations from 1996 to 2010. In columns 1 and 4, the sample is limited to fund pairs with no managers in common during quarter t . Columns 2 and 5 exclude fund pairs within the same fund family. In columns 3 and 6 the results are presented for the sample of funds where both funds in the pair are managed by a single manager. Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

Dependent Variable:	% of Overlapping Buys			% of Overlapping Sales		
	No Mgrs. Common	Different Families	Single Mgr. Pairs	No Mgrs. Common	Different Families	Single Mgr. Pairs
Sample:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Neighbors</i>	1.16 ^a (0.38)	0.83 ^b (0.36)	4.51 ^a (1.25)	0.57 ^c (0.30)	0.48 (0.30)	0.73 (0.82)
<i>SameMFCity</i>	-0.17 (0.14)	-0.14 (0.14)	0.21 (0.31)	0.22 (0.13)	0.23 ^c (0.14)	0.49 ^c (0.28)
<i>SameMediaMkt</i>	0.80 ^a (0.12)	0.73 ^a (0.12)	0.54 ^b (0.24)	0.54 ^a (0.12)	0.49 ^a (0.12)	0.14 (0.26)
<i>SameFundFam</i>	3.82 ^a (0.36)		6.26 ^a (0.64)	3.88 ^a (0.34)		5.99 ^a (0.61)
<i>MngOtherFundTogether</i>	1.91 ^a (0.45)		-0.50 (1.00)	1.43 ^a (0.51)		-0.54 (0.96)
<i>BothSmallCap</i>	0.99 ^a (0.19)	0.98 ^a (0.19)	0.79 ^a (0.28)	0.68 ^a (0.18)	0.67 ^a (0.18)	0.53 ^b (0.26)
<i>BothMidCap</i>	1.56 ^a (0.21)	1.55 ^a (0.21)	1.07 ^a (0.31)	1.14 ^a (0.18)	1.13 ^a (0.18)	0.68 ^a (0.25)
<i>BothLargeCap</i>	8.68 ^a (0.27)	8.65 ^a (0.27)	8.93 ^a (0.46)	6.82 ^a (0.24)	6.79 ^a (0.24)	7.67 ^a (0.43)
<i>BothValue</i>	2.42 ^a (0.34)	2.43 ^a (0.34)	1.77 ^a (0.66)	0.78 ^a (0.28)	0.78 ^a (0.28)	1.03 ^c (0.62)
<i>BothGrowth</i>	2.77 ^a (0.22)	2.76 ^a (0.22)	3.05 ^a (0.38)	2.43 ^a (0.21)	2.42 ^a (0.21)	2.42 ^a (0.34)
<i>BothBlend</i>	1.84 ^a (0.24)	1.85 ^a (0.24)	1.14 ^a (0.31)	1.59 ^a (0.23)	1.58 ^a (0.23)	0.94 ^a (0.30)
<i>TNAquinDiff</i>	-0.03 (0.03)	-0.03 (0.03)	-0.06 (0.05)	-0.07 ^b (0.03)	-0.07 ^b (0.03)	-0.14 ^b (0.06)
<i>TNAquinAvg</i>	0.59 ^a (0.09)	0.58 ^a (0.09)	0.58 ^a (0.17)	0.40 ^a (0.09)	0.40 ^a (0.09)	0.46 ^a (0.14)
<i>Constant</i>	1.48 ^a (0.32)	1.52 ^a (0.32)	1.39 ^b (0.57)	1.41 ^a (0.29)	1.44 ^a (0.29)	1.27 ^a (0.47)
<i>AdjR²</i>	0.14	0.14	0.16	0.10	0.10	0.12
<i>N (thousands)</i>	3,364	3,335	381	3,364	3,335	381

(a) Table III: Social interactions and mutual fund trades. Neighbor effects are visible in overlapping buys and sales.

Table IV
Extensive and Intensive Margin Overlap

The table reports the coefficient estimates and standard errors from OLS estimation of various measures of portfolio overlap on *Neighbors*, *SameMediaMkt*, and *SameMFCity*. Each row reports results for a separate regression, which include all control variables from Table II. Also reported is the average overlap in the sample for the dependent variable. Overlap in extensive purchases is the purchase overlap of newly initiated purchases. In row 1 (2) the dependent variable includes purchase overlap in stocks that were not held at the end of the previous quarter by either (at least one) fund in the pair. Intensive purchase overlap is the purchase overlap in stocks that were held by both funds during the previous quarter. Overlap in extensive sales is the sale overlap in stocks that were completely sold off during the quarter. In row 4 (5) the dependent variable includes sale overlap in stocks that are not held at the end of the current quarter by either (at least one) fund in the pair. Intensive sales overlap is the sale overlap in stocks that are held by both funds at the end of the current quarter. Note that the sum of extensive sales (buys) for at least one fund overlap and intensive sale (buy) overlap is equal to the sale (buy) overlap for each fund. The analysis in the table uses quarterly fund-pair observations for the sample of 3.4 million purchase and sale overlap observations from 1996 to 2010. The samples are limited to fund pairs that have no managers in common during quarter t . Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

Overlap in:	<i>Neighbors</i>	<i>SameMFCity</i>	<i>SameMediaMkt</i>	Avg. Overlap
Extensive buys (for both funds)	0.21 ^b (0.09)	0.10 ^a (0.04)	0.15 ^a (0.03)	1.77
Extensive buys (for at least one fund)	0.47 ^b (0.19)	0.04 (0.07)	0.50 ^a (0.06)	5.01
Intensive buys	0.68 ^a (0.21)	-0.21 ^a (0.07)	0.31 ^a (0.07)	3.47
Extensive sales (for both funds)	0.16 ^b (0.07)	0.01 (0.03)	0.01 (0.02)	1.08
Extensive sales (for at least one fund)	0.40 ^a (0.14)	0.06 (0.05)	0.15 ^a (0.05)	3.12
Intensive sales	0.18 (0.19)	0.16 (0.09)	0.39 ^a (0.08)	3.55

(b) Table IV: Extensive and intensive margin overlap. The decomposition shows whether neighbors initiate common trades or adjust common holdings.

6.5 Table V: Social Interactions and Manager Characteristics; Table VI: Social Interactions versus Preferences: Neighbor Relocations

Read: Table V shows stronger neighbor effects when managers have similar ethnic backgrounds. Same fund family and shared background also matter in some specifications. Table VI distinguishes current, future, and past neighbors. Current neighbors have strong positive overlap; future neighbors do not, while past neighbors still show some overlap in holdings.

Interpretation: Table V supports a trust or affinity channel in social transmission. Table VI is a powerful preference test: if similar residential tastes alone caused portfolio similarity, future neighbors should already

Table 9
Social Interactions and Manager Characteristics

The table reports the coefficient estimates and standard errors for OLS estimation of measures of portfolio overlap on *Neighbors* and interactions of *Neighbors* with various manager characteristics. The dependent variable for the regressions results displayed in columns 1 through 5 is the percentage of overlapping stock purchases between the funds managed by a given pair of managers, and for columns 6 through 10 the dependent variable is the percentage of overlapping stock purchases between the funds managed by a given pair of managers during the same quarter. *FundFam* is a dummy variable that is one if two funds in the fund pair belong to the same fund family. *SimilarAge* is a dummy variable that is one if the difference in the managers' ages is less than the median in the sample (10 years). *SameIndustry* is a dummy variable that is one if the sum of the managers' ages at the time they joined their current employer is less than the median in the sample (78 years). *SameCollege* is a dummy variable that is one if both managers attended the same college or university. *CareerAttendants* data are from Morningstar. *BstExpEty* (*BstExpRet*) is a dummy variable that is one if each manager has more (less) portfolio management experience than median managers in the sample. All regressions include the control variables described in Table 10. The samples are limited to manager pairs whose funds have managers in common during quarter *t*. Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by *, **, and ***, which correspond to the 1%, 5%, and 10% levels, respectively.

	% of Overlapping Holdings					% of Overlapping Buys				
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Neighbors</i>	0.88** (0.38)	1.11** (0.37)	0.99* (0.37)	0.81*	0.61 (0.34)	0.71** (0.30)	0.54*	1.03** (0.37)	0.59 (0.35)	-0.02 (0.41)
<i>Neighbors × SameFundFam</i> (1.34)						1.61** (0.64)				
<i>Neighbors × SimilarAge</i>		-0.17 (0.08)				0.43 (0.61)				
<i>Neighbors × SameIndustry</i>			3.57* (1.14)			0.07 (1.55)		2.81* (1.50)		
<i>Neighbors × SameCollege</i>				0.84 (0.79)			0.29 (0.80)		0.32 (1.00)	
<i>Neighbors × CareerAttendants</i>					-0.19 (0.18)				-0.13 (0.17)	
<i>Neighbors × BstExpEty</i>						1.16* (0.52)				3.33* (0.73)

Independent Variable:	% of Overlapping Holdings					% of Overlapping Boys				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>ghbars × dhlncsp</i>					−0.34 (0.61)					−0.57 (0.52)
<i>Exp</i>					−0.28 ^a (0.11)					−0.30 (0.12)
<i>Incsp</i>					0.28 ^a (0.13)					0.49 ^a (0.13)
<i>eFundFam</i>	3.11 ^a (0.41)	3.13 ^a (0.40)	3.15 ^a (0.40)	2.57 ^a (0.40)	3.00 ^a (0.40)	3.23 ^a (0.40)	3.29 ^a (0.40)	3.32 ^a (0.40)	2.98 ^a (0.40)	3.24 ^a (0.37)
<i>rules</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>thousands</i>	0.40 (8.823)	0.40 (8.823)	0.40 (8.823)	0.40 (8.823)	0.40 (8.823)	0.14 (7.353)	0.14 (7.353)	0.14 (7.353)	0.14 (7.353)	0.13 (5.873)

Table VI
Social Interactions versus Preferences: Neighbor Relocations

Dependent Variable:	% of Overlapping		
	Holdings (1)	Buys (2)	Sales (3)
<i>FutureNeighbors</i>	-0.38 (0.59)	-0.04 (0.56)	0.04 (0.66)
<i>Neighbors</i>	1.08 ^a (0.29)	1.10 ^a (0.37)	0.54 ^c (0.29)
<i>PastNeighbors</i>	1.18 ^b (0.49)	0.66 (0.47)	0.37 (0.43)
Controls	Yes	Yes	Yes
<i>AdjR</i> ²	0.40	0.14	0.09
<i>N</i> (thousands)	8,823	7,353	7,353

6.6 Figure 3: Portfolio overlap and manager distance; Figure 4: Portfolio overlap and neighbor tenure

11

Interpretation: These figures are difficult to reconcile with broad city-level effects. A smooth distance decay and a tenure buildup are exactly what one would expect if relationships take time and proximity raises the probability of social interaction.

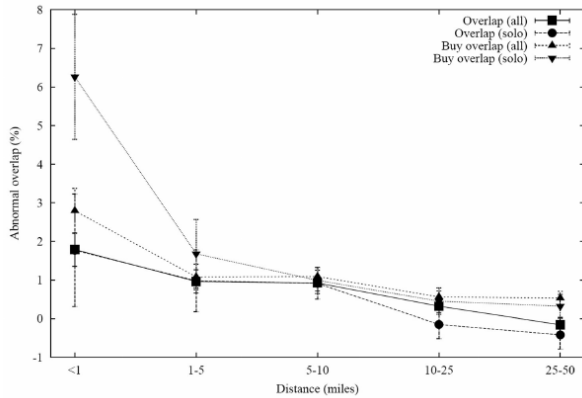


Figure 3. Portfolio overlap and manager distance. The figure plots abnormal portfolio overlap in holdings and purchases for all fund pairs and for fund pairs with single managers as a function of the distance between the residences of the funds' managers. For distances of less than five miles, we use the population density-adjusted driving distance, as explained in the text; larger distances are not adjusted and are computed using the distance between the geographic center of the cities. Abnormal overlap is the remaining portfolio overlap after controlling

(a) Figure 3: Portfolio overlap and manager distance. The neighbor effect is strongest at the closest distances and fades with distance.

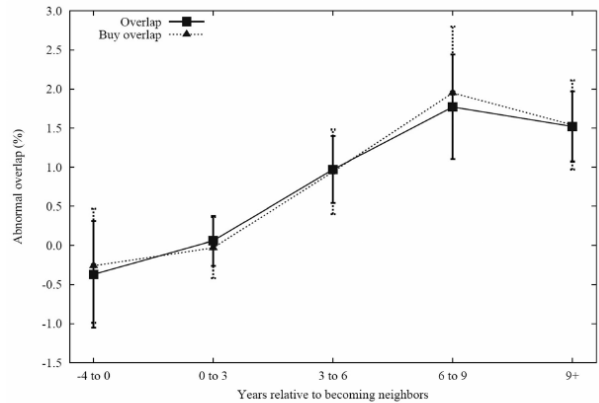


Figure 4. Portfolio overlap and neighbor tenure. The figure plots abnormal portfolio overlap in holdings and purchases as a function of time relative to a manager pair becoming neighbors. A pair is classified as becoming neighbors in the first quarter that they live within 2.6 normalized miles of each other (regardless of whether they were fund managers at the time). Specifically, we estimate

$$Overlap_{i,t} = \alpha + \beta_T FutureNeighbors_{i,t} \times I[-4 \geq NeighborsTenure_{i,t} < 0]$$

(b) Figure 4: Portfolio overlap and neighbor tenure. The effect builds after managers have lived near each other for several years.

6.7 Table VII: Social Interactions versus Preferences: Religion and Political Variables; Table VIII: Social Interactions versus Preferences: Indirect Neighbors

Read: Table VII uses proxies for local religious and political preferences. The neighbor effect remains after controlling for these preference measures. Table VIII shows that indirect neighbors - managers linked through a common neighbor but not directly adjacent - also display abnormal overlap.

Interpretation: Table VII weakens the explanation that neighbors simply share community tastes. Table VIII supports a network interpretation: information may travel through chains, not only direct pairs.

Table VII
Social Interactions versus Preferences: Additional Controls

The table reports the coefficient estimates and standard errors from OLS estimation of holdings, purchase, and sale overlap on *Neighbors* and demographic and manager control variables in Panels A and B, respectively. The analysis in the table uses the quarterly manager-pair observations outlined in Table IV. *BothReligiousAreas*, *BothJewishAreas*, and *BothCatholicAreas* are dummy variables that are one if the percentage of the population that are religious adherents, adherents to Judaism, or adherents to Catholicism in both managers' counties of residence are greater than the 75th percentile in the sample, respectively. *BothHighHomeValueAreas* (*BothLowHomeValueAreas*) is a dummy variable that is one if the median home price in both managers' zip code of residence is greater than the 75th percentile (less than the 25th percentile) in the sample. County-level religious data are from the U.S. Church Membership Data from the Association of Religious Data Archives, and zipcode-level median home prices are from the 2000 U.S. Census. *SimilarAge* is a dummy variable that is one if the difference in the managers' ages is less than the median in the sample (19 years). $HomePriceRatio = \min(HP_1, HP_2) / \max(HP_1, HP_2)$, where HP_i is the estimate of the current value of manager i 's primary residence. $LotSizeRatio = \min(LS_1, LS_2) / \max(LS_1, LS_2)$, where LS_i is the lot size of manager i 's primary residence. $HomeAgeRatio = \min(HA_1, HA_2) / \max(HA_1, HA_2)$, where HA_i is the age of manager i 's primary residence. Home characteristic data are from zillow.com. The estimates of the home prices are as of February 2013. All regressions include the control variables described in Table II. The samples are limited to manager pairs whose funds have no managers in common during quarter t . Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

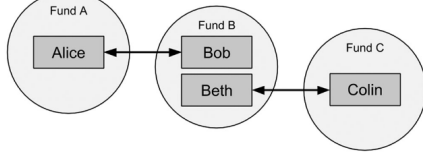
Panel A: Controlling for Demographic Characteristics						
Dependent Variable:	% of Overlapping					
	Holdings		Buys		Sales	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Neighbors</i>	1.07 ^a (0.29)	1.07 ^a (0.29)	1.10 ^a (0.37)	1.02 ^a (0.35)	0.53 ^a (0.28)	0.63 ^b (0.28)
<i>SameMFCity</i>	0.26 ^a (0.15)	0.22 (0.14)	-0.06 (0.14)	-0.11 (0.13)	0.30 ^a (0.13)	0.28 ^a (0.13)
<i>SameMediaMkt</i>	0.54 ^a (0.15)	0.11 (0.11)	0.78 ^a (0.13)	0.83 ^a (0.11)	0.56 ^a (0.13)	0.27 ^a (0.10)
<i>BothReligiousAreas</i>	0.86 ^a (0.22)	0.86 ^a (0.22)	0.59 ^a (0.17)	0.59 ^a (0.17)	0.81 ^a (0.17)	0.81 ^a (0.17)
<i>BothJewishAreas</i>	0.12 (0.21)	0.12 (0.21)	0.30 ^a (0.18)	0.30 ^a (0.19)	0.08 (0.18)	0.08 (0.18)
<i>BothCatholicAreas</i>	0.19 (0.25)	0.19 (0.25)	0.23 (0.20)	0.23 (0.20)	-0.04 (0.19)	-0.04 (0.19)
<i>BothHighHomeValueAreas</i>	-0.21 (0.22)	-0.21 (0.22)	0.14 (0.14)	0.14 (0.14)	-0.47 ^a (0.17)	-0.47 ^a (0.17)
<i>BothLowHomeValueAreas</i>	0.11 (0.22)	0.11 (0.22)	-0.24 (0.18)	-0.24 (0.18)	-0.07 (0.18)	-0.07 (0.18)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
<i>Adj R</i> ²	0.40	0.40	0.14	0.14	0.09	0.09
<i>N</i> (thousands)	8,893	8,893	7,353	7,353	7,353	7,353

(Continued)

Table VII: Religion and political preference controls, part 1. The table tests whether local preference proxies explain neighbor overlap.

Table VIII
Social Interactions versus Preferences: Indirect Neighbors

The table reports the coefficient estimates and standard errors from OLS estimation of holdings, purchase, and sale overlap on *Neighbors*, *IndirectNeighbors*, and *DistantIndirectNeighbors*. The analysis in the table uses quarterly fund-pair observations outlined in Table II. *IndirectNeighbors* is a dummy variable that is one if the funds in the pair are indirect neighbors and zero otherwise. Two funds are classified as indirect neighbors if they each have a manager who is a neighbor of a different manager of a common fund. Indirect neighbors cannot also be neighbor funds, and the minimum distance between their managers' homes is five normalized miles. For example, in the diagram below, fund A and fund C are indirect neighbors because Alice and Bob are neighbors and Beth and Colin are neighbors, and Bob and Beth comanage fund B. If, however, Alice and Colin were both neighbors of Bob, then funds A and C would not be indirect neighbors. *DistantIndirectNeighbors* is a dummy that adds the requirement that indirect neighbors be at least 50 miles apart.



All regressions include the control variables described in Table II. The samples are limited to manager pairs whose funds have no managers in common during quarter t . Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

Dependent Variable:	% of Overlapping					
	Holdings		Buys		Sales	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>IndirectNeighbors</i>	0.91 ^b (0.36)		0.80 ^b (0.39)		0.73 ^c (0.37)	
<i>DistantIndirectNeighbors</i>		1.05 ^c (0.55)		1.36 ^b (0.55)		1.18 ^b (0.52)
<i>Neighbors</i>	0.99 ^a (0.29)	0.99 ^a (0.29)	1.16 ^a (0.38)	1.16 ^a (0.38)	0.58 ^c (0.30)	0.57 ^c (0.30)
<i>SameMFCity</i>	0.19 (0.15)	0.19 (0.15)	-0.17 (0.14)	-0.17 (0.14)	0.22 (0.13)	0.22 (0.13)
<i>SameMediaMkt</i>	0.39 ^a (0.14)	0.39 ^a (0.14)	0.80 ^a (0.12)	0.80 ^a (0.12)	0.54 ^a (0.12)	0.54 ^a (0.12)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
<i>Adj R</i> ²	0.41	0.41	0.14	0.14	0.10	0.10
<i>N</i> (thousands)	4,084	4,084	3,364	3,364	3,364	3,364
Mean of dependent variable when:						
<i>Neighbors</i> = 1		11.06		11.68		8.99
<i>IndirectNeighbors</i> = 1		9.65		9.96		7.94
<i>DistantIndirectNeighbors</i> = 1		9.09		9.74		7.72

Table VIII: Indirect neighbors. The diagram and estimates show abnormal overlap for managers connected through mutual neighbor links.

Table VII—Continued

Dependent Variable:	% of Overlapping					
	Holdings		Buys		Sales	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Neighbors</i>	1.10 ^a (0.34)	1.09 ^a (0.34)	1.39 ^a (0.44)	1.35 ^a (0.44)	0.68 ^b (0.33)	0.61 ^c (0.33)
<i>SameMFCity</i>	0.36 ^b (0.15)	0.35 ^b (0.15)	0.02 (0.15)	0.01 (0.14)	0.23 ^a (0.14)	0.22 (0.13)
<i>SameMediaMkt</i>	0.71 ^a (0.17)	0.69 ^a (0.17)	0.81 ^a (0.15)	0.79 ^a (0.14)	0.69 ^a (0.16)	0.66 ^a (0.16)
<i>SimilarAge</i>		0.52 ^a (0.09)		0.40 ^a (0.08)		0.49 ^a (0.08)
<i>HomePriceRatio</i>		0.42 ^a (0.24)		0.35 ^a (0.21)		0.37 ^b (0.18)
<i>LotSizeRatio</i>		0.29 (0.19)		0.03 (0.16)		0.09 (0.16)
<i>HomeAgeRatio</i>		-0.04 (0.14)		-0.18 (0.14)		-0.07 (0.13)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
<i>Adj R</i> ²	0.40	0.40	0.14	0.14	0.09	0.09
<i>N</i> (thousands)	3,851	3,851	3,211	3,211	3,211	3,211

Table VII: Religion and political preference controls, part 2. The continuation reports additional overlap measures.

6.8 Table IX: The Performance Effect of Social Interactions: Holdings; Table X: The Performance Effect of Social Interactions: Trades

Read: Table IX studies returns to neighbor holdings. Table X studies returns to neighbor trades. The long-short portfolios based on neighbor buys versus neighbor sells deliver positive risk-adjusted performance, while corresponding non-neighbor portfolios are weaker.

Interpretation: The performance tables are central for interpreting overlap as information sharing. If neighbors merely had similar preferences, overlap would not necessarily predict risk-adjusted returns. Positive returns imply that neighbor common trades contain value-relevant signals.

Table IX

The Performance Effect of Social Interactions: Holdings

The table reports returns for “neighbor” (N) and “nonneighbor” (O) holdings of fund managers. The neighbor portfolio of fund i contains those stocks in the fund’s portfolio that are also held by at least one other fund managed by a manager who lives in the neighborhood of at least one of the managers of fund i . Holdings that do not meet this criterion comprise the nonneighbor portfolio. We average the monthly performance measures

$$R_{i,t}^N = \sum_{k \in N} \left(\frac{w_{i,k,t}}{\sum_{k \in N} w_{i,k,t}} \right) R_{k,t+1}$$

and

$$R_{i,t}^O = \sum_{k \in O} \left(\frac{w_{i,k,t}}{\sum_{k \in O} w_{i,k,t}} \right) R_{k,t+1}$$

to calculate each fund’s neighbor and nonneighbor returns in each quarter. We then average across funds in the quarter, using the dollar assets (TNA) of each fund in the previous quarter as weights, producing value-weighted average monthly returns of the neighbor and nonneighbor holdings portfolios. (N denotes the set of neighbor stocks, O the set of nonneighbor stocks, and $w_{i,k,t}$ the actual portfolio weight of fund i in stock k during quarter t .) Columns 2 and 3 report the average raw and risk-adjusted quarterly returns for the neighbor and nonneighbor portfolios for our sample period. Risk adjustment is based on DGTW benchmark returns. Column 4 reports the difference in returns between these portfolios. Standard errors are reported in parentheses. The sample is limited to fund pairs that have no managers in common during quarter t . Significance levels for tests of difference in means are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

	R^N	R^O	$R^N - R^O$
Excess return	0.62 (0.48)	0.55 (0.46)	0.07 (0.66)
DGTW (entire CRSP universe)	0.02 (0.07)	0.01 (0.08)	0.02 (0.10)
DGTW (CRSP with price > \$5)	0.03 (0.07)	-0.02 (0.08)	0.05 (0.10)

summarize the findings using the full sample of fund transactions. Column 1 of the table shows that the neighbor buy portfolio outperforms its size, book-to-market, and momentum benchmark portfolio by an average of 23 bps per month (2.8% per year). On the sell side, column 2 reveals that the neighbor

(a) Table IX: Performance effect of social interactions based on holdings. The table compares returns to positions associated with neighbor overlap.

Table X

The Performance Effect of Social Interactions: Trades

This table reports the performance of portfolios based on trades made by neighboring and nonneighboring mutual fund managers. For each fund, we form four distinct portfolios at the beginning of each quarter based on (1) whether the fund bought or sold a stock during the previous quarter and (2) whether the trade overlapped with a neighboring manager or not. Stocks that are bought are aggregated into the “buy” portfolio, while those that are sold are placed in the “sell” portfolio. We create two subgroups within the buy and sell portfolios: “neighbor” and “nonneighbor” trades. We calculate the average monthly returns of these portfolios for each fund in each quarter weighting each stock’s return in the portfolios by the new money it receives during the previous quarter. We rebalance at the end of the quarter. We then average the returns of each subportfolio across the funds in our sample using the dollar assets (TNA) of each fund in the previous quarter as weights, producing value-weighted average monthly returns of the neighbor and nonneighbor buy and sell portfolios for each quarter. Columns 1 and 2 report the risk-adjusted average quarterly returns of the neighbor buy and sell portfolios, respectively. Risk adjustment is based on DGTW benchmark returns. Columns 4 and 5 report the corresponding results for the nonneighbor portfolios. Finally, columns 3 and 6 describe the difference of the returns of the buy and sell portfolios for the neighbor and nonneighbor stocks, respectively, and column 7 provides the difference-in-difference estimate. Standard errors are reported in parentheses. Portfolio returns are reported for the entire sample period and the sample period that excludes the financial crisis, as well as for (1) all fund transactions, (2) extensive margin transactions, and (3) intensive margin transactions. The sample is limited to fund pairs that have no managers in common during quarter t . Significance levels for tests of difference in means are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

	DGTW-Adjusted Monthly Returns						
	Neighbor Portfolio			Nonneighbor Portfolio			Diff.-Diff.
	Buys	Sells	Diff.	Buys	Sells	Diff.	
All trades							
Full sample	0.23 (0.18)	-0.24 (0.19)	0.47 ^c (0.27)	0.05 (0.12)	0.05 (0.09)	0.00 (0.12)	0.48 ^c (0.27)
Excluding the financial crisis	0.33 ^c (0.19)	-0.25 (0.21)	0.58 ^c (0.30)	0.11 (0.13)	0.07 (0.10)	0.04 (0.13)	0.54 ^c (0.30)
Extensive margin trades							
Full sample	0.25 (0.23)	-0.27 (0.24)	0.53 ^c (0.29)	0.19 (0.16)	0.26 (0.18)	-0.08 (0.21)	0.60 ^c (0.35)
Excluding the financial crisis	0.30 (0.25)	-0.50 ^b (0.24)	0.81 ^a (0.29)	0.06 (0.12)	0.32 (0.21)	-0.26 (0.23)	1.07 ^a (0.34)
Intensive margin trades							
Full sample	0.32 ^c (0.18)	-0.18 (0.22)	0.50 ^c (0.27)	-0.04 (0.14)	0.04 (0.12)	-0.07 (0.12)	0.58 ^b (0.29)
Excluding the financial crisis	0.45 ^b (0.19)	-0.12 (0.24)	0.57 ^c (0.31)	0.06 (0.15)	0.07 (0.13)	-0.01 (0.13)	0.58 ^c (0.32)

Columns 4 to 6 report the corresponding results for the nonneighbor buy and sell portfolios. In contrast to our neighbor portfolios, both nonneighbor buy and sell portfolio returns are positive and very close to zero on a benchmark-adjusted basis, falling between 4 and 11 bps per month (0.5% to 1.32% per year) in the full sample and the subsample that excludes the financial crisis.

(b) Table X: Performance effect of social interactions based on trades

6.9 Table XI: Social Interactions and Stock Characteristics; Table XII: Social Interactions and Hard-to-Research Stocks

Read: Table XI asks whether coordinated neighbor trading is more common for stocks with characteristics associated with opacity. Table XII shows that neighbor coordination is especially informative for hard-to-research stocks, where information advantages should be most valuable.

Interpretation: These tables directly test the information channel. If managers share private research or informal views, the effect should be larger where public information is weaker. That is the pattern the paper

documents.

Table XI
Social Interactions, New Positions, and Stock Characteristics

The table describes correlated extensive margin buys among the fund pairs in our sample by various stock types. High (low) sales, advertising, and analyst coverage stocks are those whose corresponding characteristic is greater (less) than the median of the sample in a given year. Hard-to-research stocks are stocks that have at least one of the following criteria: low sales, low advertising, or low analyst coverage. Finally, the last few lines of the table describe shared extensive margin buys for stocks that are included and those that are not included in the S&P 500 index, respectively. Column 1 shows the mean probability that, if fund i acquires a new position in stock k in quarter t , a nonneighbor fund that does not hold the stock in quarter $t-1$ also acquires a new position in stock k in quarter t or $t+1$. Column 2 restricts nonneighbor funds to those located within the same media market as fund i . Column 3 provides the corresponding average probabilities for neighbors. The difference in probabilities between columns 2 and 1 is reported in column 4, while the difference between columns 3 and 2 is shown in column 6. Finally, columns 5 and 7 summarize the percent difference in neighbor and nonneighbor probabilities. The average probabilities reported in the table represent time-series averages of the corresponding quarterly frequencies. The sample is limited to fund pairs that have no managers in common during quarter t . Significance levels for tests of difference in means are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

<i>SameMediaMkt =</i> <i>Neighbors =</i>	Prob. of Establishing New Positions Together When			Diff.	Avg. % Diff. in Prob.	Diff.	Avg. % Diff. in Prob.
	0	1	1				
	(1)	(2)	(3)				
				(3)-(1)		(3)-(2)	
All stocks	3.39 (0.13)	4.45 (0.14)	5.26 (0.22)	1.77* (0.17)	52.30* (5.50)	0.91* (0.13)	19.71* (2.75)
High sales stocks	3.23 (0.11)	4.00 (0.12)	4.68 (0.21)	1.45* (0.14)	44.40* (4.54)	0.68* (0.12)	15.49* (2.56)
Low sales stocks	1.42 (0.07)	1.97 (0.13)	2.78 (0.24)	1.39* (0.23)	106.92* (20.67)	0.81* (0.19)	45.79* (14.16)
Difference					-62.47* (20.56)		-30.21* (14.92)
High advertising stocks	3.49 (0.14)	4.36 (0.16)	4.80 (0.25)	1.31* (0.16)	37.10* (4.66)	0.54* (0.16)	11.68* (3.67)
Low advertising stocks	1.86 (0.08)	2.49 (0.14)	3.48 (0.34)	1.62* (0.33)	87.37* (19.45)	0.99* (0.29)	31.40* (11.28)
Difference					-50.27* (19.09)		-19.72* (10.93)

(Continued)

Table XI: Social interactions and stock characteristics, part 1. The table examines common new positions by stock visibility and research difficulty.

Table XII
The Performance Effect of Social Interactions: Trades in Hard-to-Research Stocks

This table reports the performance of portfolios based on trades made in hard-to-research and easy-to-research stocks by neighbor and nonneighbor mutual fund managers. Stocks are categorized as "hard-to-research" if they satisfy at least one of the following criteria: low sales, low advertising expense, or low analyst coverage, where these variables follow the definitions described in Table XI. For both hard- and easy-to-research stocks we form four distinct portfolios for each fund-quarter by following the methodology outlined in Table X: a neighbor buy, neighbor sell, nonneighbor buy, and nonneighbor sell portfolio. We then average the returns of each subportfolio in each quarter across the funds in our sample using the dollar assets (TNA) of each fund in the previous quarter as weights, producing value-weighted monthly average returns for each quarter and each subportfolio. Columns 1 and 2 report the risk-adjusted average returns of the hard-to-research buy and sell neighbor and nonneighbor portfolios for our entire sample period, respectively. Risk adjustment is based on DGTW benchmark returns. Columns 4 and 5 report the corresponding results for the easy-to-research portfolios. Columns 3 and 6 describe the difference of the returns of the buy and sell portfolios within the neighbor and nonneighbor portfolios for hard-to-research and easy-to-research stocks, respectively, and column 7 provides the difference-in-difference estimate between the hard-to-research stocks and easy-to-research stocks within the neighbor and nonneighbor portfolios. The third row of the table shows the difference-in-difference between the neighbor and nonneighbor portfolios within hard-to-research and easy-to-research stocks. The sample is limited to fund pairs that have no managers in common during quarter t . Standard errors are reported in parentheses. Significance levels for tests of difference in means are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

	DGTW-Adjusted Monthly Returns						
	Hard-to-Research Stocks			Easy-to-Research Stocks			Diff.-Diff.
	Buy	Sell	Diff.	Buy	Sell	Diff.	
Neighbor portfolio	0.35 (0.58)	-1.64 ^b (0.82)	1.99 ^b (0.90)	0.24 (0.18)	-0.15 (0.17)	0.39 (0.26)	1.60 ^c (0.85)
Nonneighbor portfolio	-0.61 ^c (0.38)	-0.25 (0.40)	-0.36 (0.35)	0.09 (0.12)	0.10 (0.09)	-0.01 (0.12)	-0.35 (0.35)
Diff.-Diff.			2.35 ^b (0.90)			0.40 (0.26)	1.95 ^b (0.86)

(a) Table XII: Neighbor trading and hard-to-research stocks. The performance advantage is concentrated among opaque stocks.

6.10 Figure 5: Bootstrap Analysis

Read: Figure 5 plots the distribution of the neighbor coefficient from 5,000 bootstrap simulations that impose the null of no neighbor effect by randomizing neighbor assignments within media markets. The actual coefficient lies far in the right tail.

Interpretation: The bootstrap shows that the measured neighbor effect is not an artifact of the large sample or of random spatial clustering inside media markets. It is much larger than what random neighbor assignment would produce.

Table XIII—Continued

<i>SameMediaMkt =</i> <i>Neighbors =</i>	Prob. of Establishing New Positions Together When			Diff.	Avg. % Diff. in Prob.	Diff.	Avg. % Diff. in Prob.
	0	1	1				
	(1)	(2)	(3)				
				(3)-(1)		(3)-(2)	
High analyst coverage stocks	3.21 (0.11)	3.99 (0.12)	4.67 (0.21)	1.47* (0.14)	45.68* (4.15)	0.69* (0.12)	15.80* (2.78)
Low analyst coverage stocks	1.46 (0.21)	2.15 (0.24)	3.32 (0.42)	1.84* (0.31)	96.86* (22.46)	0.87* (0.28)	41.86* (14.16)
Difference					-50.98* (21.56)		-26.06* (13.84)
Easy-to-research stocks	3.32 (0.11)	4.33 (0.15)	5.22 (0.21)	1.91* (0.17)	60.63* (5.75)	0.90* (0.13)	20.37* (3.02)
Hard-to-research stocks	1.91 (0.15)	2.59 (0.18)	3.66 (0.27)	1.76* (0.21)	105.09* (12.9)	1.07* (0.18)	43.41* (7.44)
Difference					-44.46* (13.92)		-25.00* (7.57)
S&P 500 stocks	4.11 (0.15)	5.09 (0.18)	5.73 (0.20)	1.62* (0.22)	39.23* (5.60)	0.65* (0.19)	11.62* (3.87)
Non-S&P 500 stocks	2.25 (0.09)	2.91 (-0.11)	3.33 (0.30)	1.08* (0.16)	50.44* (7.56)	0.42* (0.15)	14.10* (4.70)
Difference					-11.20 (8.75)		-2.48 (5.65)

Table XI: Social interactions and stock characteristics, part 2. The continuation reports additional opacity categories.

Table XIII
Neighbor Trading and Past Performance

The table reports the coefficient estimates and standard errors from the OLS estimation of holdings, purchase, and sale overlap on *Neighbors*, *PosPastDGTWPerf*, and the interaction of the two. *PosPastDGTWPerf* is a dummy variable that is one if the DGTW-adjusted return of the portfolio created from lagged overlapping purchases minus sales of the fund pair is greater than zero, and zero otherwise. The analysis uses quarterly fund-pair observations for the sample of fund pairs that had positive overlap in the past. The regressions include the control variables described in Table II. The sample is limited to fund pairs whose funds have no managers in common during quarter t . Standard errors, two-way clustered by each fund in the pair, are in parentheses. Significance levels are denoted by a, b, c, which correspond to the 1%, 5%, and 10% levels, respectively.

Dependent Variable:	% of Overlapping	
	Holdings (1)	Buy (2)
<i>Neighbors</i>	0.97 ^a (0.37)	1.23 ^a (0.46)
<i>Neighbors</i> × <i>PositivePastDGTWPerf</i>	0.45 ^b (0.21)	0.54 ^c (0.31)
<i>PositivePastDGTWPerf</i>	0.09 ^b (0.04)	-0.07 ^c (0.04)
Controls	Yes	Yes
$AdjR^2$	0.40	0.14
N (thousands)	1,699	1,671

IV. Robustness

A. Alternative Overlap Measures

We perform a number of robustness tests to confirm that our results are not sensitive to a particular measure of overlap. We estimate our baseline models from columns 2 and 6 of Table III and column 5 of Table II using a number of different overlap measures.

First, as an alternative to the holdings measure used to estimate the regres-

(b) Table XIII: Social interactions and past performance. The table tests whether past performance explains neighbor overlap.

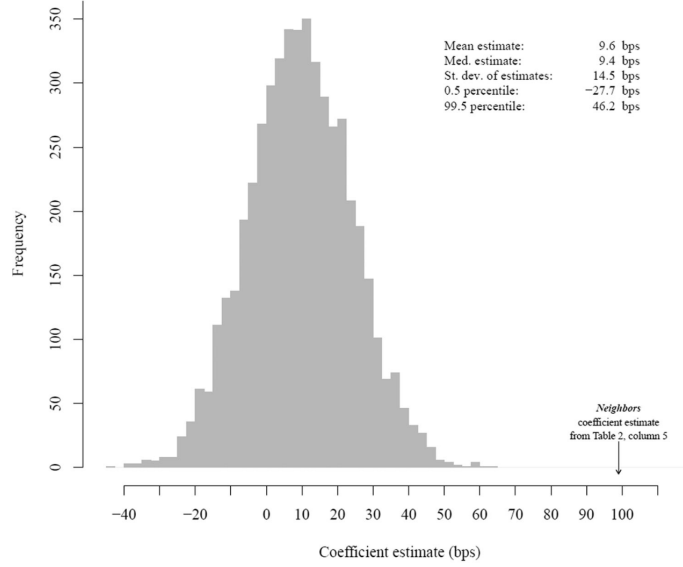


Figure 5. Bootstrap analysis. The figure shows the distribution of *Neighbors* coefficient estimates from 5,000 bootstrap simulations of the regression equation in Table II, column 5. We impose the null of no neighbor effect by randomizing who is a neighbor. If a manager-pair observation is in the same media market (*SameMediaMkt* = 1), we randomly assign the pair to be neighbors with probability 3.15%, which gives the same overall proportion of neighbors as in the sample.

Figure 5: Bootstrap analysis. The realized neighbor coefficient is compared with a null distribution from randomized neighbor assignments.

7 Short summary

- The paper studies whether professional fund managers transmit investment ideas through local social networks.
- It uses residential address histories to identify fund managers who live near one another.
- Neighbor managers have more similar portfolio holdings and trades than managers who only share a city or media market.
- The effect is stronger for purchases than for sales, consistent with idea sharing.
- The effect is stronger among managers with similar ethnic backgrounds.
- Future-neighbor placebo tests do not explain the results, while current and past neighbor effects suggest relationship formation and persistence.
- The effect decays with distance and rises with neighbor tenure.
- Indirect-neighbor results suggest that information can travel through social chains.
- Neighbor buy-sell portfolios earn positive abnormal returns, especially in hard-to-research stocks.
- The evidence supports value-relevant social transmission among professional investors, not just shared local preferences.

8 Conclusion

8.1 Core conclusion

Pool, Stoffman, and Yonker show that social interactions shape the portfolios and trades of professional mutual fund managers. Residential neighbors hold and trade more similar stocks, and the overlap is not explained by broad city-level media markets, fund styles, same fund families, or common manager structures.

8.2 Economic mechanism

The mechanism is informal information transmission. Managers who live near one another are more likely to meet, develop trust, and exchange ideas. The evidence is strongest for purchases, grows with neighbor tenure, is amplified by ethnic similarity, appears through indirect connections, and is concentrated in hard-to-research stocks.

8.3 Why the identification is persuasive

The key strength is the contrast between social proximity and community proximity. Same city and same media market capture broad local information, while the neighbor variable captures a narrower interaction channel. Future-neighbor and bootstrap tests further weaken the preference and random-correlation interpretations.

8.4 Limitations

The paper does not directly observe conversations, friendship, or the content of information exchanged. Residential proximity is a proxy for relationship formation. Some unobserved local sorting may remain. In addition, neighbor trades represent a small part of fund portfolios, so the overall effect on total fund performance may be modest despite large returns in specific hard-to-research trades.

8.5 Takeaway

The paper adds an important social-network layer to delegated asset management. Even among sophisticated professionals, investment ideas diffuse through informal peer networks. Geography matters not only because firms and investors are local, but because people who make portfolio decisions interact socially.